

HYPER-3

Matthew Reda · axiom

Author Note

The HYPER-3 study evaluates which of three daily doses should go into phase III and whether any dose harms participants. Across the age strata, 400 adults with elevated or stage-1 hypertension were randomized 2:1:1:1 to standard of care or one of three daily doses, with 100 randomized in band age 25 to 35, 160 in band age 36 to 50, and 140 in band age 51 plus. The target effect of arm on change is identified via the backdoor route, yielding pooled contrasts against control of -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg) for 10 mg, -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg) for 20 mg, and -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg) for 40 mg evaluated against a decision threshold of 0.00 mmHg. For the 40 mg dose, the contrast against control is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg) in band age 25 to 35 and 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg) in band age 51 plus. Across the trial, 372 units contributed to the primary endpoint, and 0.797 of units remained on study at the final week.

Abstract

Four hundred adults with elevated or stage-1 hypertension were randomized to evaluate three daily doses against standard of care. The trial addressed which daily dose should advance to phase III and whether any dose harms the participants. The effect of treatment on blood pressure change is identified via the backdoor route identified in the identification graph. The strongest standing assumption is that dropout is ignorable given baseline. The pooled contrasts for 40 mg, 20 mg, and 10 mg each settled below the decision threshold on the favourable side. For the 40 mg dose in participants aged 51 and older, the contrast settled above the threshold on the unfavourable side at 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg), whereas the same dose settled below the threshold in participants aged 25 to 35. These findings indicate that while pooled estimates support advancing the doses, heterogeneity across age strata reveals an unfavorable increase in blood pressure for older participants on the 40 mg dose.

Reported quantities: 40 mg vs control, pooled; 20 mg vs control, pooled; 10 mg vs control, pooled; 40 mg vs control, age 51+

Introduction

This report addresses which of three daily doses should go into phase III, and whether any of them is harming the people taking it. The analysis answers this question using 40 mg versus control, pooled. The target quantity is identified. The decision turns on a decision threshold of 0.00 mmHg. An interval that falls entirely on one side of 0.00 mmHg settles the comparison. An interval spanning 0.00 mmHg leaves the question unsettled. Such an interval does not mean there is no effect.

Methods

To address the question established in the introduction, 400 adults with elevated or stage-1 hypertension were randomized 2:1:1:1 to standard of care or to one of three daily doses, stratified by age band, with the design parameters recorded in Table 1. The protocol estimates three contrasts against a shared control rather than one dose against another. The target is the effect of arm on change, which is identified via the backdoor route shown in Figure 1. This identification turns on the arrows from adherence to change, age to change, arm to adherence, arm to change, and baseline to change. Seated systolic pressure was measured weekly, and both endpoints average the readings inside a window—weeks 5–8 for safety and weeks 9–12 for the primary endpoint—rather than reading a single visit. Averaging several correlated readings roughly halves the variance of a unit's endpoint, which is the difference between a safety boundary that can fire inside a stratum and one that cannot. Each dose was compared with control by ANCOVA on the change from baseline, adjusting for baseline pressure, and adjusting for age band when the contrast is pooled across bands. Randomization licenses the intent-to-treat contrast and nothing else; adjusting for a pre-randomization covariate buys precision at no cost, whereas adjusting for anything measured afterwards would forfeit the identification the randomization provided. Twelve harm contrasts—each dose against control overall and inside each age band—were monitored at scheduled safety reviews against a posterior-probability boundary with a 2 mmHg clinical margin. Pre-specifying the bands as monitored contrasts makes harm within a single age band visible before the trial ends, which no estimator applied at closeout would have found earlier. The analysis relies on two unresolved assumptions, named dropout is ignorable given baseline and the pooled contrast answers the question. What each of these assumptions claims is stated in the limitations section.

Protocol

arm_i ~ blocked randomization within age band, allocation
2:1:1:1

Analysis

change_i = beta0 + tau * treated_i + beta1 * baseline_i + eps_i
pooled, add: + sum_b gamma_b * 1[age_band_i = b]
change_i = mean(SBP_i, weeks 9-12) - SBP_i(baseline)
tau = the contrast reported; treated_i = 1 for the dose arm

Safety monitoring

stop for harm at look k when
 $P(\tau_g > \delta \mid \text{data}_k) > b_k$ for any monitored contrast g

Δ = the clinical margin; b_k = the boundary at look k

The identification graph

From	Arrow	To	Kind
adherence	→	change	direct effect
age	→	change	direct effect
arm	→	adherence	direct effect
arm	→	change	direct effect
baseline	→	change	direct effect

Table 1. The identification graph, as arrows

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
dropout is ignorable given baseline	units who left the study differ from those who stayed only through what was measured before randomization	is assumed and has not been checked	a tipping-point analysis over plausible departures from ignorability
the pooled contrast answers the question	a single number over all age bands describes the effect the decision is about	is known to fail	heterogeneity across the pre-specified strata

Table 2. Standing assumptions

assumption Adherence was not adjusted for: it lies on the path from arm to outcome, and conditioning on it would break the randomization.

assumes *no_conditioning_on_post_randomization_variables* (satisfied): the analysis conditions on nothing measured after assignment; adherence is a mediator and dropout is a descendant of the outcome

assumption The pooled 40 mg contrast is reported alongside the per-band contrasts, never instead of them.

assumes *the_pooled_contrast_answers_the_question* (violated): a single number over all age bands describes the effect the decision is about

Assumption ledger

Results

The comparison of active doses against control addresses the question and decision threshold specified in the introduction under the identification strategy from the methods. For 10 mg versus control, the pooled estimate is -5.27 mmHg (90% WALD -6.63 to -3.91 mmHg). For 20 mg versus control, the pooled estimate is -4.80 mmHg (90% WALD -6.17 to -3.44 mmHg). For 40 mg versus control, the pooled estimate is -1.53 mmHg (90% WALD -2.97 to -0.10 mmHg). These estimated quantities appear alongside their intervals and thresholds in Table 2. When evaluated within age categories, the estimate for 40 mg versus control in the age 25-35 group is -5.26 mmHg (90% WALD -8.17 to -2.36 mmHg). For 40 mg versus control in the age 51+ group, the estimate is 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg). These estimated quantities and their intervals are displayed in Figure 2 relative to the decision threshold.

40 MG VS CONTROL, POOLED	20 MG VS CONTROL, POOLED	10 MG VS CONTROL, POOLED	40 MG VS CONTROL, AGE 51+	40 MG VS CONTROL, AGE 25-35
-1.53 mmHg	-4.80 mmHg	-5.27 mmHg	6.20 mmHg	-5.26 mmHg
[-3.0, -0.1] (90% WALD)	[-6.2, -3.4] (90% WALD)	[-6.6, -3.9] (90% WALD)	[4.0, 8.4] (90% WALD)	[-8.2, -2.4] (90% WALD)

Model checking

Diagnostic checks assess the behavior of the sample and model fit rather than the estimated effect. A check of this kind can lower confidence in an estimate and can never establish one. The recorded diagnostics for HYPER-3 are reported in Table 3. The fraction of units still on study at the final week was 0.797. The study includes 372 units contributing to the primary endpoint. Across age categories, there were 100 units randomized in band age 25 35. There were 160 units randomized in band age 36 50. The study recorded 140 units randomized in band age 51 plus. These metrics examine observed retention and sample partitions, whereas the broader assumptions specified in the methods remain unexamined by these diagnostic counts.

UNITS STILL ON STUDY AT THE FINAL WEEK	UNITS CONTRIBUTING TO THE PRIMARY ENDPOINT	UNITS RANDOMIZED IN BAND AGE 25 35	UNITS RANDOMIZED IN BAND AGE 36 50	UNITS RANDOMIZED IN BAND AGE 51 PLUS
0.797	372	100	160	140

Discussion

For 40 mg vs control in the age 51+ group, the estimate settled above the threshold on the unfavourable side, marking the primary divergence in these data. This result contrasts with the younger cohort, where 40 mg vs control for ages 25-35 settled below the threshold on the favourable side. The pooled comparisons mask this age variation: 40 mg vs control, 20 mg vs control, and 10 mg vs control each settled below the threshold on the favourable side when pooled. That divergence at 40 mg indicates that the favourable pooled finding does not extend to older participants. Because the analysis relies on the backdoor identification route established in the methods, these findings license reading the estimates as effects of the treatments rather than simple associations. They do not license inferences beyond what that identification structure supports. All interpretations remain conditional on 2 unresolved assumptions. The limitations section details what each of those assumptions claims. What was found is that while younger cohorts and lower pooled doses yield favourable outcomes, the highest dose produces an unfavourable outcome in older participants, raising the question of how phase III dose selection should manage age-specific risks.

Conclusions

No single daily dose can be recommended for phase III across all groups, as the 40 mg dose harms older participants. For participants aged 51 and older, the 40 mg versus control contrast settled on the unfavourable side of the decision threshold at 6.20 mmHg (90% WALD 3.98 to 8.42 mmHg). Conversely, the pooled comparisons for 10 mg, 20 mg, and 40 mg versus control each settled on the favourable side of the threshold. The subgroup contrast for 40 mg versus control among participants aged 25 to 35 likewise settled on the favourable side. Because the subgroup results diverge across age brackets, a pooled figure averages benefit with harm. The effect is identified following the structure established in the methods. This answer is conditional on 2 unresolved assumptions stated in the limitations section. Specifically, the conclusions hold provided that dropout is ignorable given baseline and that the pooled contrast answers the question.

Limitations

The analysis relies on standing assumptions, beginning with one that is violated. The assumption that the pooled contrast answers the question claims that a single number over all age bands describes the effect the decision is about. This assumption licenses using an aggregate contrast to address the decision problem set out in the introduction. That claim is challenged by heterogeneity across the pre-specified strata. In addition, the assumption that dropout is ignorable given baseline remains unverified. This assumption claims that units who left the study differ from those who stayed only through what was measured before randomization. It licenses the identified effect despite subject attrition under the design described in the methods. This condition is challenged by a tipping-point analysis over plausible departures from ignorability.

Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
40 mg vs control, pooled	pooled_40	-1.53 mmHg (90% WALD -2.97 to -0.10 mmHg)	identify.ols (ANCOVA)
20 mg vs control, pooled	pooled_20	-4.80 mmHg (90% WALD -6.17 to -3.44 mmHg)	identify.ols (ANCOVA)
10 mg vs control, pooled	pooled_10	-5.27 mmHg (90% WALD -6.63 to -3.91 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 51+	oldest_40	6.20 mmHg (90% WALD 3.98 to 8.42 mmHg)	identify.ols (ANCOVA)
40 mg vs control, age 25-35	youngest_40	-5.26 mmHg (90% WALD -8.17 to -2.36 mmHg)	identify.ols (ANCOVA)
Units still on study at the final week	retention	0.797	—
Units contributing to the primary endpoint	analysed	372	—
Units randomized in band age 25 35	n_age_25_35	100	—
Units randomized in band age 36 50	n_age_36_50	160	—
Units randomized in band age 51 plus	n_age_51_plus	140	—

Table 3. Quantities and their sources

Field	Value
endpoint	mean change, weeks 9-12

interval_mass	90%
seed	20260821
world	nbs/case-studies/hypertension/hyper3.py
evidence hash	f6c953a837c79459cb14b031868abf7f0b1adb6e235b3 2abca19246ffd9d5ecc
narration: abstract	gemini-3.5-flash-lite — verified
narration: title_page	gemini-3.7-flash — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified
narration: limitations	gemini-3.7-flash — verified

Table 4. Run

Tables

Table 1. The design, as recorded parameters.

Step	Parameter	Value
Identification	graph	adherence -> change, age -> change, arm -> adherence, arm -> change, baseline -> change
Identification	graph hash	d02f82660e65
Identification	route	backdoor
Identification	status	identified
Protocol	allocation	2:1:1:1
Protocol	arms	standard_of_care, dose_10, dose_20, dose_40
Protocol	enrollment weeks	16
Protocol	follow-up weeks	24
Protocol	primary endpoint week	12
Protocol	strata	age_25_35, age_36_50, age_51_plus
Protocol	units randomized	400
Measurement	minimum visits in window	3
Measurement	retention at final week	79.8%
Analysis	estimator	ANCOVA (OLS on change, baseline-adjusted)
Analysis	interval mass	90%
Analysis	pooled covariates	baseline_centred, is_age_36_50, is_age_51_plus
Analysis	within-stratum covariates	baseline_centred
Safety monitoring	clinical margin (mmHg)	2
Safety monitoring	monitored contrasts	12

Table 2. Estimated quantities with their intervals and thresholds.

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
40 mg vs control, pooled	-1.53	mmHg	-2.97 to -0.10 (90% WALD)	0.00	below threshold
20 mg vs control, pooled	-4.80	mmHg	-6.17 to -3.44 (90% WALD)	0.00	below threshold
10 mg vs control, pooled	-5.27	mmHg	-6.63 to -3.91 (90% WALD)	0.00	below threshold
40 mg vs control, age 51+	6.20	mmHg	3.98 to 8.42 (90% WALD)	0.00	above threshold
40 mg vs control, age 25-35	-5.26	mmHg	-8.17 to -2.36 (90% WALD)	0.00	below threshold

Table 3. Recorded diagnostics.

Check	Value	Unit	Interval
Units still on study at the final week	0.797	—	—
Units contributing to the primary endpoint	372	—	—
Units randomized in band age 25 35	100	—	—
Units randomized in band age 36 50	160	—	—
Units randomized in band age 51 plus	140	—	—

Figures

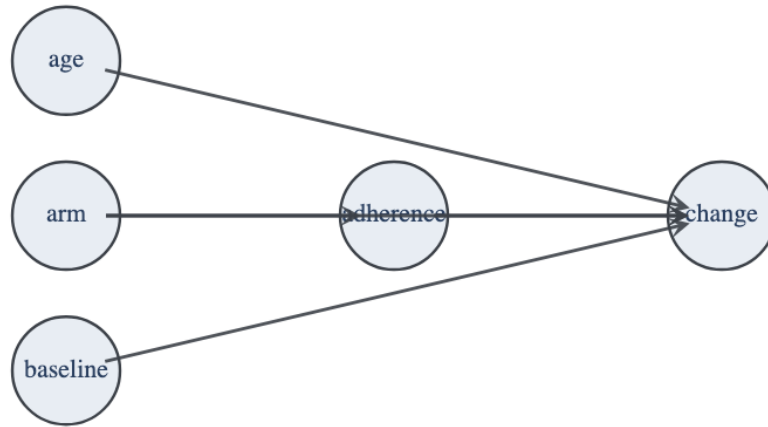


Figure 1. The identification graph. A hollow node was not measured and a dashed edge is an unmeasured common cause.

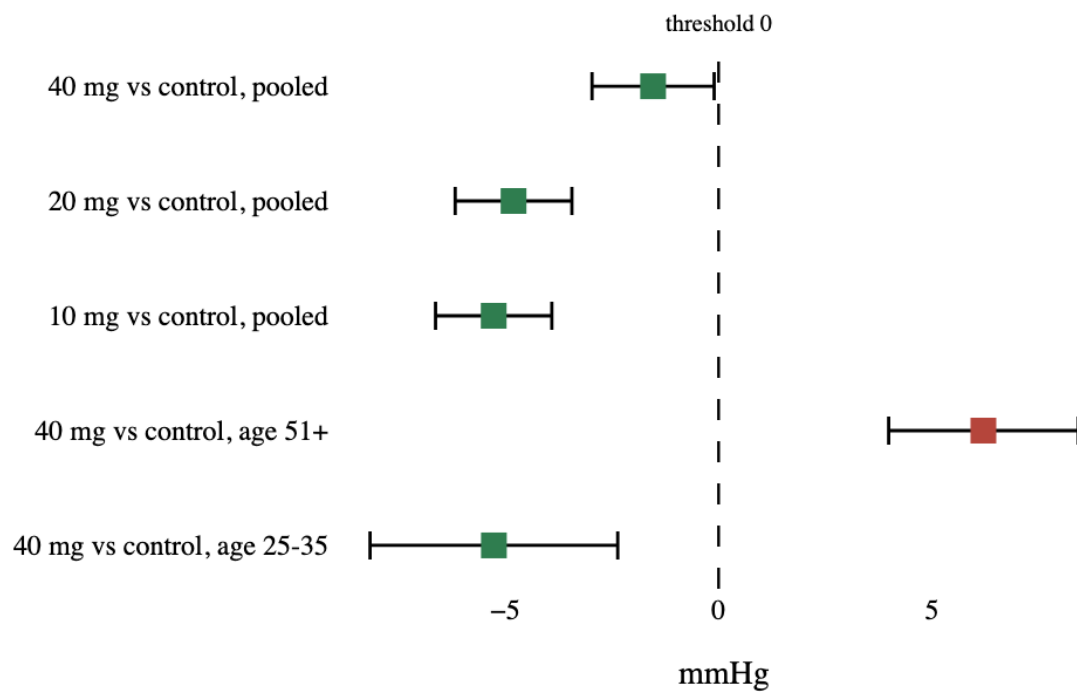


Figure 2. Estimated quantities with their intervals. The dashed rule marks the decision threshold; a row whose interval crosses it is unsettled rather than null.